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Emotion Experience and Expression Goals Shape Emotion Regulation Strategy Choice

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Research has begun to investigate how goals for emotion experience—how people want to feel—influence the selection of emotion regulation strategies to achieve these goals. We make the case that it is not only how people want to *feel* that affects strategy selection, but also how they want to be *seen* to feel. Incorporating this expressive dimension distinguishes four unique emotion goals: (1) to experience and express emotion; (2) to experience but not express emotion; (3) to express but not experience emotion; and (4) to neither experience nor express emotion. In six experiments, we investigated whether these goals influenced choices between six common emotion regulation strategies. Rumination and amplification were selected most often to meet Goal 1—to experience and express emotion. Expressive suppression was chosen most often to meet Goal 2—to experience but not express emotion. Amplification was chosen most often to meet Goal 3—to express but not experience emotion. Finally, distraction was chosen most often to meet Goal 4—to neither experience nor express emotion. Despite not being chosen most for any specific goal, reappraisal was the most commonly selected strategy overall. Our findings introduce a new concept to the emotion goals literature and reveal new insights into the process of emotion regulation strategy selection.

Keywords: emotion goals, emotion regulation, emotion experience, emotion expression

Supplemental materials: https://doi.org/10.1037/emo0001012.supp

Emotions are helpful: They can feel good, help people connect, stay motivated, and perform well. Emotions are also harmful: They can feel bad, alienate people, overwhelm them, and cloud their judgment. Because of this changing utility, people sometimes want to be more emotional and at other times want to be less emotional. People tune their emotions in this way through the process of emotion regulation, in which they change a current emotional state into a desired one (Gross et al., 2011). But what governs *how* people choose to regulate their emotions? There is a large literature on the consequences of emotion regulation, but researchers know much less about the antecedents of emotion regulation and the factors that influence *which* emotion regulation strategies will be chosen under *what* circumstances.

An emerging perspective seeks to answer these questions through the lens of emotion goals—the planned destination of the emotion regulation journey. Until recently, research on emotion goals and strategies has occurred largely in parallel (Tamir, 2016), but recent work suggests that how people want to feel influences which strategies they choose (e.g., Millgram et al., 2019). We broaden this perspective to argue it is not only how people want to *feel* that influences the strategies they choose, but how they want others to *see* them feel. This expressive dimension has been neglected in the literature on emotion goals to date, but is necessary in order to understand the full spectrum of desired emotional states. Uniting methods from social and cognitive psychology, we test whether goals for emotion experience and expression influence how people choose which emotion regulation strategy to use.

Emotion Goals

Goals are the engine of human striving, structuring decisions and behaviors that range from the mundane (which outfit to wear)

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All data and code, along with supplementary detail and analyses, as well as preregistrations for the final two experiments, are available on the OSF (https://osf.io/ndkrq/).

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to the profound (which cause to fight for). Within this broad field, researchers distinguish between the goals people have (e.g., lose weight) and the means by which they achieve those goals (e.g., diet, exercise; Gollwitzer, 1993). This means that one goal may be pursued using a range of different strategies.

As with the broader goals literature, emotion researchers distinguish between emotion goals and the emotion regulation strategies used to pursue those goals (e.g., Eldesouky & Gross, 2019; Mauss & Tamir, 2014; Millgram et al., 2019; Millgram & Tamir, 2019; Tamir, 2016). Emerging work on emotion goals focuses on how people want to *feel*, although some research has also investigated how people want to *appear* to feel (Tamir, 2016; Tsai, 2007). These represent goals for emotion experience and goals for emotion expression, respectively.

Emotion experience goals refer to the emotions that people aim to feel (Mauss & Tamir, 2014). Most existing research on this topic has focused on hedonic goals, which aim to maximize positive emotion or minimize negative emotion (Riediger et al., 2009; Tamir, 2016). However, a comparatively smaller body of research shows that people also hold contra-hedonic emotion experience goals, which aim to maximize negative emotion or minimize positive emotion. These goals are often held for instrumental reasons, such as feeling anger in order to win an argument or feeling anxiety in order to facilitate performance on an important task (e.g., Tamir & Ford, 2009, 2012; Tamir et al., 2008). Regardless of the emotional direction, this work shows that people aim to experience emotion in strategic ways.

Emotion expression goals refer to the emotions that people aim to show on their faces (Greenaway & Kalokerinos, 2019). Although not always referred to as emotion expression goals per se, research suggests that people sometimes deliberately aim to show, or hide, expressions of particular emotions. This is the message of literatures on display rules (Ekman, & Friesen, 1975), emotional labor (Grandey, 2000; Hochschild, 1979, 1983), and contextually appropriate expression and suppression of emotion (Ansfield, 2007; Kalokerinos et al., 2014; Schall et al., 2016; Szczurek et al., 2012). These literatures demonstrate that people often aim to express emotion to meet strategic ends.

Although some existing research has investigated goals for emotion experience and expression separately, a new perspective advocates for studying these goals in concert, arguing that people hold goals for the experience and expression of emotion simultaneously (Greenaway & Kalokerinos, 2019; Osborne et al., 2020). Specifically, experience and expression goals can be aligned or misaligned, resulting in four distinct emotion goals: to (1) experience and express emotion, (2) experience but not express emotion, (3) express but not experience emotion, and finally (4) neither experience nor express emotion (see Figure 1).

This conceptualization distinguishes what one wants to *feel* (experience goals) from what one wants to *show* (expression goals). Sometimes, goals for the dimensions are in alignment (e.g., wanting to feel and show), but other times goals for the dimensions may be in opposition (e.g., wanting to feel but not show). Considered holistically, Goals 1 and 4 align goals for emotion experience and expression, while Goals 2 and 3 set goals for emotion experience and expression in opposition. Early work shows that people hold these kinds of "two-dimensional" goals—goals in which people aim to achieve a certain level of emotion experience *and* expression—in addition to holding "one-dimensional" goals for either emotion experience *or* emotion expression in isolation (Greenaway & Kalokerinos, 2019).

This two-dimensional perspective captures the richness of emotional life, both as it is experienced and expressed. Focusing on only experience goals *or* expression goals risks missing the variety of reasons why, and directions in which, people aim to adjust their emotional state. Emotion experience and expression are modestly correlated at .24 to .42 (Gross et al., 2000), and thus goals to target one dimension (e.g., reduce felt anxiety) may go hand in hand with goals to target the other dimension (e.g., reduce shown anxiety), but they may just as easily be in opposition (e.g., to feel pride but avoid showing it to others).

This two-dimensional perspective on emotion goals offers an opportunity to better understand the factors that predict which emotion regulation strategies people choose to meet certain emotion goals. Because different strategies influence emotion experience and expression in different ways, certain strategies will be more or less tailored to achieve a specific emotion goal (e.g., Millgram et al., 2015; Millgram et al., 2019). To test these ideas, we investigated whether, and how, goals for emotion experience and expression shape the emotion regulation strategies that people choose.

Emotion Regulation

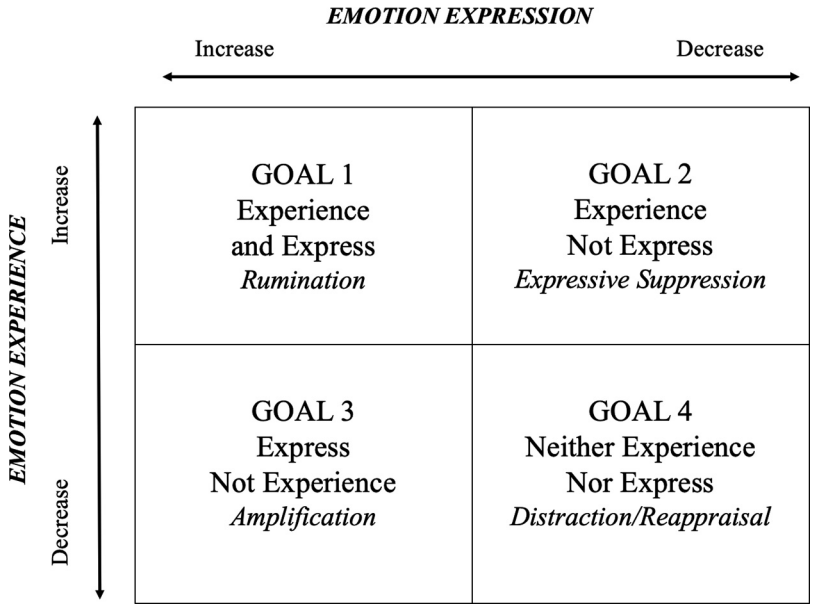
Emotion regulation refers to the means by which people influence which emotions they experience and express (e.g., Gross, 1998b). Within a broad literature, the dominant theoretical lens is the (extended) process model of emotion regulation, which delineates emotion regulation into stages ranging from initial decisions to regulate (identification), to choices about how to regulate (selection), and the process and outcomes of actioning those choices (implementation; Gross, 2015). Research has focused more on implementation and its emotional aftermath than on the identification or selection stages (cf., Sheppes et al., 2011; Sheppes et al., 2014). In this work, we explore how emotion goals shape strategy selection.

The original process model identified several strategies people may select to regulate their emotions (Gross, 1998a). Similar strategies are grouped into categories that include: (a) situation selection, (b) situation modification, (c) attentional deployment, (d) cognitive change, and (e) response modulation. Of these, the latter three families have received the most research attention (Gross & Thompson, 2007) and appear to be used in daily life (Brans et al., 2013). We focus on six emotion regulation strategies subsumed within these three categories, which are by far the most studied in the emotion regulation literature (e.g., Webb et al., 2012). Here, we summarize the general impact these strategies appear to have on emotion experience and expression when considered across the literature.[1]

Attentional deployment refers to strategies in which people direct their attention to engage with, or disengage from, an emotional

[1] Although we aim to give a birds-eye view of the impact of these strategies on emotion experience and expression, it is important to note that the actual selection and implementation of these strategies does differ across people and contexts (e.g., Dixon-Gordon et al., 2015). Indeed, this is the message of a broader literature on regulatory flexibility that argues for the utility of selecting the "right" strategy for the "right" situation (Bonanno & Burton, 2013).

Figure 1
Goals for Emotion Experience and Expression



Note. Theoretical predictions about which emotion regulation strategies may best facilitate these goals are represented in italics.

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stimulus (Gross & Thompson, 2007). These alternate options describe two well-researched emotion regulation strategies: rumination and distraction. *Rumination* is a regulation strategy in which people focus their attention on their emotions or the cause of their emotions (Nolen-Hoeksema, 2000). Although typically investigated in the context of negative emotion, some research has studied positive rumination (often referred to as "savoring"; Bryant & Veroff, 2007). For both valences, research suggests that rumination increases emotion experience (Erisman & Roemer, 2010; Quoidbach et al., 2010; Webb et al., 2012) and emotion expression (Webb et al., 2012).

In comparison, *distraction* is a strategy in which people actively divert cognitive attention from an emotional stimulus (e.g., checking one's phone while watching a scary film; Gross, 1998b). In a meta-analysis, Webb et al. (2012) found that certain types of distraction decreased both emotion experience and expression. Less researched is the impact of distraction on positive emotion, but the little work that has been done suggests that it also reduces positive emotion experience and expression (Quoidbach et al., 2010).

Cognitive change is characterized by shifting one's interpretation of an emotion stimulus (Gross & Thompson, 2007). The most commonly researched regulation strategy of this sort is *cognitive reappraisal*, which involves changing one's perspective about an emotional stimulus to alter its emotional impact (Gross, 2002; Ochsner et al., 2004). Studies indicate that certain types of reappraisal, such as distancing, can reduce positive and negative emotion experience (Kalokerinos et al., 2015) and emotion expression (Webb et al., 2012).

Response modulation occurs when people seek to change their emotional response. One strategy of this sort is *experiential suppression*, sometimes termed experiential avoidance, in which people attempt to diminish or repress an emotional response (Chawla & Ostafin, 2007). Because it is broadly defined and may represent a suite of behavioral phenomena rather than a specific strategy, there is relatively limited evidence for the impact of experiential suppression per se on emotion experience and less on emotion expression. That said, some research does suggest this strategy can paradoxically increase emotion experience (Farach et al., 2008; Hayes et al., 2004; Kashdan et al., 2014). Another response modulation strategy is *expressive suppression*, which is an attempt to hide the outward display of emotional behavior (Gross & John, 1998). Given this definition, it is no surprise that suppression reduces positive and negative emotion expression (Gross & Levenson, 1997). Perhaps more surprising is that research shows suppression does not affect positive or negative emotion experience in the short term relative to control (Kalokerinos et al., 2015).

In contrast to suppression, *amplification*, also known as expressive exaggeration, refers to exaggerated expression of emotion (Gross, 1998a). Comparatively fewer studies have investigated amplification, but the research that has been done suggests that amplification increases emotion expression, but does not appear to change emotion experience (Demaree et al., 2004; Demaree et al., 2006; Kunzmann et al., 2005).

Emotion Regulation to Meet Emotion Goals

To summarize the impact of these emotion regulation strategies on emotion experience and expression, in general, *rumination* increases emotion experience and expression, *expressive suppression* decreases emotion expression but leaves emotion experience unchanged, and *amplification* increases emotion expression but leaves emotion experience unchanged. *Distraction* and certain types of *reappraisal* decrease emotion experience and expression. This provides a loose road map for understanding how different

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strategies may assist people in reaching a particular emotional destination.

As outlined in Figure 1, we mapped these five emotion regulation strategies onto specific emotion goals. Note that because the strategies appear to affect positive and negative emotion in similar ways, this mapping holds for both valences. Because it increases emotion experience and expression, rumination seems well placed to meet the goal to *experience and express* emotion (Goal 1 in Figure 1). Expressive suppression decreases emotion expression but leaves emotion experience unchanged, recommending it as a strategy to meet the goal to *experience but not express* emotion (Goal 2). In contrast, amplification increases emotion expression but leaves experience unchanged, and thus may assist in meeting the goal to *express but not experience* emotion (Goal 3). Finally, distraction and certain types of reappraisal decrease emotion experience and expression, and thus could be chosen to achieve the goal to *neither experience nor express* emotion (Goal 4). It was unclear to us which goal experiential suppression would meet, given the specific emotional and behavioral consequences of this strategy are relatively underresearched in the experimental literature.

Although theory is one way to derive predictions about which emotion regulation strategies people may choose, another way is to observe actual emotion regulation choice. This approach is at the center of a literature that investigates selection of different emotion regulation strategies and the conditions that shape such choices (e.g., Sheppes et al., 2011; Sheppes et al., 2014). This emerging literature examines emotion regulation as it is revealed through people's volitional behavior.

Research has begun to explore the link between emotion goals and strategy selection, focusing exclusively on goals for emotion experience. For example, Eldesouky and English (2019) assessed prohedonic experience goals (e.g., goals to experience more positive emotion) and contra-hedonic experience goals (e.g., goals to experience more negative emotion) and their relationships with strategy use, both globally and in daily life. Endorsement of prohedonic goals predicted greater use of antecedent-focused strategies such as reappraisal and distraction (see also English et al., 2017). These findings are correlational, but experimental research manipulating goals to increase or decrease emotion experience has also found that people choose specific strategies to meet these goals. Specifically, Millgram et al. (2019) compared distraction and reappraisal, and found that people instructed to hold a goal to increase emotion experience in a picture-viewing task selected rumination on a majority of trials, while people instructed to hold a goal to decrease emotion experience selected distraction on a majority of trials. These choices appear relatively adaptive, given that rumination reliably increases, and distraction reliably decreases, emotion experience.

Research to date therefore suggests that people are capable of choosing strategies that help them to achieve specific emotion experience goals. However, experimental work has focused on assessing choice between two strategies, such as distraction and reappraisal (e.g., Sheppes et al., 2014) or distraction and rumination (e.g., Millgram et al., 2019). Moreover, this work has focused exclusively on emotion experience goals, such that emotion expression goals, and the interface between experience and expression, remain unexplored. To address these issues, we investigated the experimental impact of emotion experience and expression goals on choice of a wide range of emotion regulation strategies: In addition to distraction, rumination, and reappraisal, we also assessed amplification, and expressive and experiential suppression. In combining top-down theory driven and bottom-up revealed choice approaches, we are able to determine whether actual choice of emotion regulation strategies differs from what theory would predict is the most effective strategy to meet an emotion goal.

The Present Research

This research represents the first attempt to systematically map the impact of emotion experience and expression goals on emotion regulation strategy choice. We conducted six experiments using a mixture of retrospective designs in which people recalled personal emotion goals, and lab designs in which people were assigned to hold specific emotion goals. For each study pair, we conducted one experiment on positive emotion and one on negative emotion. All data and code, along with supplementary detail and analyses, as well as preregistrations for the final two experiments, are available on the OSF.

Experiments 1a and 1b asked participants to recall a situation in which they held one of four emotion goals and report on the strategy they chose to meet that goal. Experiments 2a and 2b assessed emotion regulation choice in the lab after participants were experimentally assigned to hold a specific emotion goal. Experiments 3a and 3b (preregistered) used a discrete choice paradigm to assess not only which strategies are *chosen* while holding a specific emotion goal, but also which strategies are *avoided* under the same conditions. This permits stronger inferences about the strategies that are most and least useful for meeting specific emotion goals. The paradigm and corresponding modeling analysis deconfounds the strategies people believe are most valuable in meeting emotion goals from individual differences in the cognitive processes people use to choose a strategy. Together, the studies paint a more nuanced picture of goal-directed emotion regulation choice.[2]

Experiment 1a: Positive Emotion

Method

Participants

The initial sample was 189 MTurk workers from the United States who were paid $1 for their participation. We aimed to recruit approximately 200 unique participants to Experiment 1a and 1b (striving for ∼50 per between-subjects cell before exclusions). We did not conduct an a priori power analysis. Instead, our Bayesian analysis evaluates how much information the actual data provide, rather than how consistently a result may occur in hypothetical future replications of the experiment, as in frequentist power analyses. So, for a Bayesian analysis, a small sample size translates to posterior distributions with greater uncertainty than larger sample sizes. This means we are less likely to observe

[2] In Experiments 1a–2b we also asked participants to self-report on strategy effectiveness. Because our focus is on emotion regulation choice, we did not analyze these data, but we include these variables in the datasets on OSF to make such analyses possible in future.

differences between posterior distributions (conditions), which biases our approach toward the null for small sample sizes. We removed 14 participants who provided an irrelevant or nonsensical description of the event they were recalling, and 65 participants who failed to recall their goal condition correctly in the comprehension check. Thus, our final sample was 110 participants (M_{age} = 37.53, SD_{age} = 11.41, 47 women, 63 men).

Design

We employed a 2 (emotion expression goal: express vs. not express) × 2 (emotion experience goal: experience vs. not experience) between-subjects design.

Procedure and Materials

Emotion Goal Manipulation. Participants were randomly assigned to one of the four conditions. They were asked to recall a situation in their lives in which they held the emotion goal that matched the condition that they were assigned to. The text read as follows (with text that differed by condition in italics in parentheses):

> Please think about a situation in your life in which you wanted to [*BOTH experience AND express/experience BUT NOT express/express BUT NOT experience/NEITHER experience NOR express*] positive emotion. That is, a situation in which you wanted to [*both feel positive emotion and show positive emotion/feel positive emotion BUT NOT show positive emotion/show positive emotion BUT NOT feel positive emotion/neither feel positive emotion NOR show positive emotion*].

Participants were asked to provide a brief description of this situation. A coder read through these descriptions and noted any descriptions that did not make sense or were irrelevant to the prompt, which were excluded from analysis.

Emotion Regulation Strategy Choice. To assess strategy choice, participants were asked "If you encountered this situation again, which of the following strategies would you choose to use in order to meet your goal to [*both feel and show/feel but not show/show but not feel/neither feel nor show*] positive emotion?"]. They were then asked to select one strategy from a list of six emotion regulation strategies adapted and extended from Brans et al. (2013) and Kalokerinos et al. (2019). These strategies were cognitive reappraisal ("change my perspective in order to change my positive emotion"), rumination ("focus or dwell on my positive emotion"), expressive suppression ("suppress the outward expression of my positive emotion"), distraction ("distract myself from the event or my positive emotion"), experiential suppression ("suppress the inner experience of my positive emotion"), and amplification ("emphasize the outward expression of my positive emotion"). Participants were also provided with an option to select "other" if they would have chosen something different from the strategies above, but no participant selected this option in Experiment 1a.

Comprehension Check. This check was designed to identify participants who could not recall the goal to which they were assigned. Participants were asked "What type of situation did you recall at the time of this survey?" and could choose from six options: the four goal conditions, as well as "not sure/do not know" and "none of the above." We excluded participants who did not correctly select their goal condition.

Analysis

We analyzed the emotion regulation strategy choices with a multinomial logistic regression, also known as the multinomial logit (MNL) model. The outcome variable was the chosen strategy across participants. The predictor variables were the six emotion regulation strategies. These predictors lead to six corresponding regression coefficients that can be interpreted in terms of *utility*, where utility corresponds to the relative subjective value of an option; this interpretation is standard in the discrete choice literature where MNLs are widely applied to data of the form studied here (Train, 2009). In our context, utility is interpreted as the relative value that participants place on each of the six emotion regulation strategies in meeting their goal. A strategy with higher utility is more likely to be chosen.

We estimated the utility parameters of the MNL model in a Bayesian framework. We provide an overview of the estimation details in the main text and report complete details in supplementary materials available on the OSF. The analysis generates a posterior distribution for the six utility parameters. The posterior distribution gives the range of feasible values for the true value of the utility parameter, given the observed data and the structure of the MNL model. From the posterior distributions we compute the "posterior predictive *p*-value" for pairwise comparisons between the six strategies. In our context, the posterior predictive *p*-value is interpreted as the probability that one strategy has greater utility than another strategy (i.e., is more likely to be chosen). We conducted targeted hypothesis tests with the posterior predictive *p*-value for each of the emotion goals. Each goal has a strategy theory would predict is likely to be particularly effective in achieving the goal:

1. BOTH experience AND express: rumination
2. Experience BUT NOT express: expressive suppression
3. Express BUT NOT experience: amplification
4. NEITHER experience NOR express: reappraisal or distraction

For each goal, we report the posterior predictive *p*-value that the hypothesized emotion regulation strategy has higher utility than the alternative strategies. We use thresholds such that a posterior predictive *p*-value $> .95$ indicates the hypothesized strategy had reliably *higher* utility than an alternative strategy, and a posterior predictive *p*-value $< .05$ indicates the hypothesized strategy had reliably *lower* utility than an alternative strategy. Posterior predictive *p*-values between .05 and .95 are indeterminate, meaning there was insufficient information in the data to determine whether the two utility parameters were different.

Ethics

The experiments received ethical clearance from the University of Queensland (17-PSYCH-4–06-JH) and the University of Melbourne (1953768.1).

Results and Discussion

Figure 2 shows the posterior distribution of the utility parameters for the four goal conditions in all experiments,

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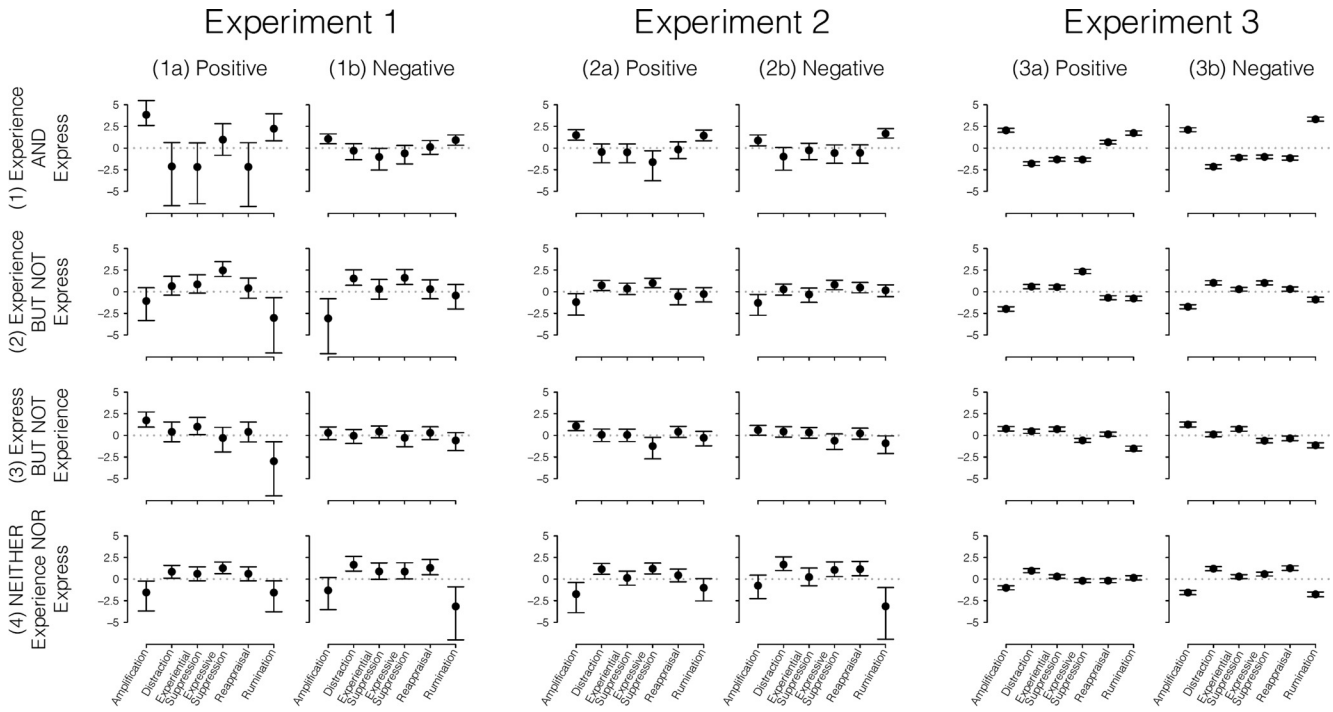
Figure 2

Posterior Distribution of the Utility Parameters for the Three Experiments (Columns), Four Goal Conditions (Rows), and Six Emotion Regulation Strategies (x-Axis)



Note. Dots and bars show the posterior median and 95% credible interval, respectively. The dotted gray line is a reference point showing zero utility (neither positive or negative utility). Positive utility indicates a strategy is more likely to be chosen and negative utility indicates a strategy is less likely to be chosen to meet a particular emotion goal.

including Experiment 1a. Table 1 displays the posterior predictive *p*-values for a given goal in all experiments, including Experiment 1a.

For Goal 1 (experience and express emotion), choices were somewhat consistent with our theoretical expectation in that rumination was chosen more often than most strategies, but it was also chosen *less* often than amplification. As expected theoretically, for Goal 2 (experience but not express emotion), expressive suppression was chosen more than all other strategies. Also somewhat consistent with theoretical expectations, Goal 3 (express but not experience emotion), amplification was chosen more than all strategies except experiential suppression. Somewhat unexpectedly, for Goal 4 (neither experience nor express emotion), distraction and reappraisal were chosen more than rumination and amplification, but not more than experiential and expressive suppression.

Broadly speaking, people were united in their strategy selection for Goal 2 to experience but not express emotion: Participants preferred expressive suppression—a strategy shown to dampen the expression but not experience of emotion—over other strategies. A plurality of participants also selected amplification to meet Goal 3—to experience but not express emotion—suggesting that people may have a relatively clear idea of how to meet goals that distinguish emotion expression from experience. People were divided between amplification or rumination to meet Goal 1—to experience and express emotion—and thus may have been attempting to target the expressive and experiential dimensions separately with two strategies. There was much less clarity in choice regarding Goal 4—to neither experience nor express emotion—with participants divided between several strategies.

Experiment 1b: Negative Emotion

Method

Participants

We initially recruited 211 MTurk workers from the United States who were paid $1 for their participation. We removed two participants who specified "other" rather than selecting an emotion regulation strategy from the list provided, 14 participants who provided an irrelevant or nonsensical description of the event they were recalling, and 63 participants who failed the comprehension check. Thus, our final sample was 123 participants (M_{age} = 34.45, SD_{age} = 10.56, 70 women, 53 men).

Design, Procedure, Materials, and Analysis

These were identical to Experiment 1a, except all references to positive emotion were replaced with negative emotion.

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Table 1
Posterior Predictive p-Values That the Hypothesized Strategy Has a Higher Utility Than Alternative Strategies Across Experiments

		Experiment 1		Experiment 2		Experiment 3	
Emotion goal	Strategy comparison	(1a) Positive	(1b) Negative	(2a) Positive	(2b) Negative	(3a) Positive	(3b) Negative
(1) Experience AND Express	Rumination vs. expressive suppression	.94	**>.99**	**>.99**	**>.99**	**>.99**	**>.99**
	Rumination vs. experiential suppression	**>.99**	**>.99**	**>.99**	**>.99**	**>.99**	**>.99**
	Rumination vs. amplification	**<.001**	.35	.43	**.99**	**.04**	**>.99**
	Rumination vs. distraction	**>.99**	**.99**	**>.99**	**>.99**	**>.99**	**>.99**
	Rumination vs. reappraisal	**>.99**	**.95**	**>.99**	**>.99**	**>.99**	**>.99**
(2) Experience NOT Express	Expressive suppression vs. rumination	**>.99**	**>.99**	**>.99**	.93	**>.99**	**>.99**
	Expressive vs. experiential suppression	**>.99**	**.99**	**.95**	**.99**	**>.99**	**>.99**
	Expressive suppression vs. amplification	**>.99**	**>.99**	**>.99**	**>.99**	**>.99**	**>.99**
	Expressive suppression vs. distraction	**>.99**	.59	.78	.89	**>.99**	.48
	Expressive suppression vs. reappraisal	**>.99**	**>.99**	**>.99**	.78	**>.99**	**>.99**
(3) Express NOT Experience	Amplification vs. rumination	**>.99**	.90	**>.99**	**>.99**	**>.99**	**>.99**
	Amplification vs. expressive suppression	**>.99**	.83	**>.99**	**.99**	**>.99**	**>.99**
	Amplification vs. experiential suppression	.94	.39	**.99**	.75	.59	**.99**
	Amplification vs. distraction	**.99**	.72	**.99**	.66	.93	**>.99**
	Amplification vs. reappraisal	**.99**	.50	**.96**	.81	**>.99**	**>.99**
(4) NEITHER Experience NOR Express	Distraction vs. rumination	**>.99**	**>.99**	**>.99**	**>.99**	**>.99**	**>.99**
	Distraction vs. expressive suppression	.15	**.97**	.43	**.95**	**>.99**	**>.99**
	Distraction vs. experiential suppression	.69	**.97**	**.99**	**>.99**	**>.99**	**>.99**
	Distraction vs. amplification	**>.99**	**>.99**	**>.99**	**>.99**	**>.99**	**>.99**
	Reappraisal vs. rumination	**.99**	**>.99**	**.97**	**>.99**	**.05**	**>.99**
	Reappraisal vs. expressive suppression	.06	.81	**.03**	.59	.52	**>.99**
	Reappraisal vs. experiential suppression	.51	.81	.72	**.96**	**.01**	**>.99**
	Reappraisal vs. amplification	**>.99**	**>.99**	**>.99**	**>.99**	**>.99**	**>.99**

Note. A large p (>.95), shown in bold, indicates the theoretically expected strategy was chosen more than the comparator; a small p (<.05), shown in bold, indicates the hypothesized strategy was chosen less than the comparator.

Results and Discussion

Figure 2 shows the posterior distribution of the utility parameters for the four goal conditions in all experiments, including Experiment 1b. Table 1 displays the posterior predictive p-values for a given goal in all experiments, including Experiment 1b.

For Goal 1 (experience and express emotion), rumination was chosen more than all other strategies except amplification. For Goal 2 (experience but not express emotion), expressive suppression was chosen more than all other strategies except distraction. For Goal 3 (express but not experience emotion), amplification was not chosen more than any other strategy. For Goal 4 (neither experience nor express emotion), distraction was chosen more than every other strategy and reappraisal was chosen more often than rumination and amplification, but not experiential and expressive suppression.

As in Experiment 1a, a plurality of participants chose expressive suppression to meet Goal 2 to experience but not express emotion. Unlike in Experiment 1a, there was no clear strategy chosen to meet Goal 3 to experience but not express emotion. As in Experiment 1a, participants were divided between amplification and rumination for Goal 1—to experience and express emotion—strategies that may be directed at targeting different emotion dimensions. Again there was little clarity in strategy selection regarding Goal 4—to neither experience nor express emotion—although distraction was chosen more than other strategies.

These results provide initial insight into strategy selection to meet emotion goals that align or misalign desired levels of experience and expression. Strategy choice varied considerably across goals and across the studies, with little consistency aside from expressive suppression being chosen most to meet Goal 2 to experience but not express emotion, and amplification and rumination chosen most to meet Goal 1 to experience and express emotion.

These studies are among the first to experimentally investigate two-dimensional emotion goal antecedents of emotion regulation choice among a range of strategies. However, the experiments produced imprecise posterior distributions for the utility parameters due to a larger than expected number of exclusions, and relied on retrospective recall of emotion goals. Experiments 2a and 2b addressed these issues, manipulating two-dimensional emotion goals in the lab and recruiting a larger number of participants per cell.

Experiment 2a: Positive Emotion

Method

Participants

We recruited 200 Australian participants (M_{age} = 20.43, SD_{age} = 4.24, 155 women, 45 men). Participants were either first-year psychology undergraduates who received course credit for their participation, or community participants recruited through a paid participant database who received AU$10 for participation.

Design

We employed a 2 (emotion expression goal: express vs. not express) × 2 (emotion experience goal: experience vs. not experience) between-subjects design.

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Procedure and Materials

Participants were tested individually in the lab. After providing informed consent, they reported on baseline emotional experience, watched the positive emotion induction film clip, and then reported postinduction emotional experience.

Participants were randomly assigned to one of the four goal conditions and received the emotion goal manipulation. They were told that they would enact the emotion goal manipulation during a recording in which they discussed the film clip just watched. Participants were asked to choose which emotion regulation strategy they would use to meet that goal (prevideo choice). They then recorded the video, which took about 30 s, after which participants specified the strategy that they actually used while recording the video (postvideo choice).

Positive Emotion Induction. We induced positive emotion using a short film clip from *Finding Nemo* which was validated in Kalokerinos et al. (2015).

Emotion Goal Manipulation. Participants were randomly assigned to one of the four conditions. They were told that they would be recording a short video clip describing what happened in the emotion induction film. The text read as follows (with text that differed by condition in italics and parentheses):

> It is very important when recording this video that you try to [*BOTH experience AND express/experience BUT NOT express/express BUT NOT experience/NEITHER experience NOR express*] positive emotion. That is, while recording the video please try to [*both feel positive emotion and show positive emotion/feel positive emotion BUT NOT show positive emotion/show positive emotion BUT NOT feel positive emotion/neither feel positive emotion NOR show positive emotion].

To direct the video response, participants were asked to answer two questions during the video. First, they were asked to describe the main characters and what happened in the clip. Second, they were asked to explain whether they would recommend the video to another person, and why or why not.

Emotional Experience. We adapted an emotional experience scale from Posner et al. (2005), and asked participants to report directly before and after the emotion induction the degree to which they felt contented, happy, amused, proud, hopeful, and joyful (positive emotion, $\alpha_{preinduction} = .82$; $\alpha_{postinduction} = .82$), and the degree to which they felt sad, nervous, angry, fearful, irritated, and stressed (negative emotion, $\alpha_{preinduction} = .75$; $\alpha_{postinduction} = .83$). Participants responded to these items on a 7-point scale (1 = *not at all,* 7 = *very much*).

Emotion Regulation Choice. Participants completed the same measure of emotion regulation strategy choice as Experiment 1, both prevideo (indicating the strategy they thought they would use) and postvideo (indicating the strategy they actually used). We found that most participants specified the same strategy in both the prevideo and postvideo choice. We conducted a sensitivity analysis and confirmed that the results did not differ when analyzing prevideo choice. As a result, we report here only the results with the postvideo choice variable.

Behavioral Expression. As a behavioral goal fulfilment check for emotion expression, we ran the videos generated by participants through iMotions (2013), a software package which coded for specific instances of positive and negative emotion expression. The output was the frequency of positive and negative emotion expressions detected by the software. Analyses were performed using 195 participants because five participants experienced technical errors or withdrew consent for their videos to be analyzed.

Self-Reported Experience and Expression. As a self-reported goal fulfilment check for emotion experience and expression in the different goal conditions, participants reported how much positive emotion (contented, happy, amused, proud, hopeful, and joyful) they *experienced* ($\alpha = .85$) and *expressed* ($\alpha = .89$) while recording the video (scale adapted from Posner et al., 2005). Participants also reported how much negative emotion (sad, nervous, angry, fearful, irritated, and stressed) they *experienced* ($\alpha = .78$) and *expressed* ($\alpha = .78$) while recording the video. Participants responded to these items on a 7-point scale (1 = *not at all,* 7 = *very much*). We report the effects of the goal manipulation on self-reported emotion experience below, and the effects on self-reported emotion expression in supplementary materials.

Analysis

This was identical to Experiment 1a, except the outcome variable was postvideo choice.

Results and Discussion

Emotion Induction

A paired-samples t-test revealed the emotion induction was successful: participants reported more positive emotion after watching the clip ($M = 4.02$, $SD = 1.14$) than before ($M = 3.70$, $SD = 1.08$), $t(199) = 5.07$, $p < .001$, $d = .36$.

Self-Reported Emotion Experience

A 2 (emotion expression goal: express vs. not express) × 2 (emotion experience goal: experience vs. not experience) between-subjects ANOVA was applied to self-reported emotion experience while recording the video. This checked the degree to which participants reported being able to fulfil their experience goal.

There was a significant main effect of experience goal on positive emotion experience during video recording, $F(1, 196) = 4.53$, $p = .035$, $d = .29$, such that participants reported experiencing more positive emotion in the *experience* condition ($M = 3.48$, $SD = 1.25$) compared with the *do not experience* condition ($M = 3.12$, $SD = 1.19$). There was also a significant main effect of express goal on positive emotion experience, $F(1, 196) = 9.95$, $p = .002$, $d = .44$, such that participants reported experiencing more positive emotion in the *express* condition ($M = 3.57$, $SD = 1.26$) compared with the *do not express* condition ($M = 3.04$, $SD = 1.14$). There was no significant interaction, $F(1, 196) = .30$, $p = .582$.

Behavioral Emotion Expression

A 2 (emotion expression goal: express vs. not express) × 2 (emotion experience goal: experience vs. not experience) between-subjects ANOVA revealed a significant main effect of express goal on the frequency of computer-coded positive emotion expressions during video recording, $F(1, 191) = 57.73$, $p < .001$, $d = 1.08$, such that participants spent a greater percent of the time expressing positive emotion in the *express* condition ($M = 32.59$, $SD = 27.79$) compared with the *do not express* condition ($M =$

8.42, SD = 14.84). There was no significant main effect of experience goal on behavioral positive emotion expression, $F(1, 191) = 3.70$, $p = .056$, and no significant interaction, $F(1, 191) = .57$, $p = .451$.

Goal Utility

Figure 2 shows the posterior distribution of the utility parameters for the four goal conditions in all experiments, including Experiment 2a. Table 1 displays the posterior predictive p-values for a given goal in all experiments, including Experiment 2a.

For Goal 1 (experience and express emotion), rumination was chosen more than all other strategies except amplification. For Goal 2 (experience but not express emotion), expressive suppression was chosen more than all other strategies except distraction. For Goal 3 (express but not experience emotion), amplification was chosen more often than any other strategy. For Goal 4 (neither experience nor express emotion), distraction was chosen more than all strategies except expressive suppression, although reappraisal was only chosen more often than rumination and amplification and *less* often than expressive suppression.

Strategy selection was relatively clear on asymmetric goals. As in earlier experiments, a plurality of participants chose expressive suppression to meet Goal 2—to express but not experience emotion—although distraction was also a popular strategy for this goal. Amplification was chosen more than other strategies to meet Goal 3—to experience but not express emotion. Once again, amplification and rumination were equally chosen to meet Goal 1—to experience and express emotion—and a clear choice did not emerge to meet Goal 4—to neither experience nor express emotion although distraction was again commonly chosen.

Inspection of the goal fulfilment checks suggested that participants may have had difficulty translating these goals into desired emotional states: Participants instructed to express positive emotion also reported experiencing more positive emotion regardless of the emotion experience goal they were assigned. This may indicate that people have difficulty separating experiential and expressive dimensions of emotion; a possibility made more likely by the fact that emotion experience and expression correlate positively with one another, although the correlation is typically only small to medium in size (Gross et al., 2000).

Experiment 2b: Negative Emotion

Method

Participants

We recruited 200 Australian participants ($M_{\text{age}} = 20.93$, $SD_{age} = 2.63$, 129 women, 70 men, one did not specify). Participants were either first-year psychology undergraduates who received course credit for their participation, or community participants recruited through a paid participant database who received AU$10 for participation.

Design, Procedure, Materials, and Analysis

These were identical to Experiment 2a, except all references to positive emotion were replaced with negative emotion. In this experiment we induced negative emotion using a short film clip from *The Lion King* which was validated in Kalokerinos et al. (2015).

Results and Discussion

Emotion Induction

A paired-samples t-test revealed emotion induction succeeded: Participants reported experiencing greater negative emotion after watching the clip ($M = 2.76$, $SD = 1.06$) than before ($M = 2.17$, $SD = 1.07$), $t(199) = 6.11$, $p < .001$, $d = .43$.

Self-Reported Emotion Experience

There was no significant main effect of experience goal on negative emotion experience during video recording, $F(1, 196) = .16$, $p = .687$, no significant main effect of express goal, $F(1, 196) = 1.05$, $p = .306$, and no significant interaction, $F(1, 196) = .18$, $p = .670$.

Behavioral Emotion Expression

There was a significant main effect of express goal on the frequency of computer-coded negative emotion expressions during video recording, $F(1, 193) = 4.01$, $p = .046$, $d = .28$, such that participants spent a greater percent of the time expressing negative emotion in the *express* condition ($M = 7.94$, $SD = 14.35$) compared with the *do not express* condition ($M = 4.39$, $SD = 10.50$). There was no significant main effect of experience goal on behavioral negative emotion expression, $F(1, 193) = 1.27$, $p = .261$, and no significant interaction, $F(1, 193) = 1.55$, $p = .215$.

Goal Utility

Figure 2 shows the posterior distribution of the utility parameters for the four goal conditions in all experiments, including Experiment 2b. Table 1 displays the posterior predictive p-values for a given goal in all experiments, including Experiment 2b.

For Goal 1 (experience and express emotion), rumination was chosen more often than any other strategy. For Goal 2 (experience but not express emotion), expressive suppression was chosen more often than experiential suppression and amplification, but not more than other strategies. For Goal 3 (express but not experience emotion), amplification was chosen more often than rumination and expressive suppression, but not more than other strategies. For Goal 4 (neither experience nor express emotion), distraction and reappraisal were chosen more often than any other strategy, except expressive suppression in the case of reappraisal.

Results of Experiment 2b were less clear than other experiments, except in the case of Goal 1—to experience and express emotion—for which rumination was chosen consistently above other strategies. For all other goals, no clearly preferred strategy emerged that was in line with what theory would predict. This may in part be due to the fact that some participants had difficulty fulfilling their emotion goals—participants instructed to experience emotion did not feel more negative than participants instructed to not experience emotion.

A further limitation of this and the preceding experiments is they assessed strategy selection as a one-off choice that did not capture variation across trials. To address this, in our final two experiments we used a paradigm from cognitive psychology that

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asks participants to distinguish between the *best* and *worst* choice in a given array. This method has been used to discriminate between preference for, or avoidance of, different options in the domains of consumer choice, health-related decisions, and policy preferences (e.g., see Hawkins et al., 2019; Jones et al., 2015; Louviere et al., 2015). Applying this paradigm to assess emotion regulation choice allowed us to introduce multiple trials which, in addition to increasing the power and reliability of our tests, helps to triangulate not only the strategies people prefer to use to meet a certain emotion goal, but also the strategies they prefer to avoid under the same conditions.

Experiment 3a: Positive Emotion

Method

This study was preregistered: https://aspredicted.org/da9yw.pdf.

Participants

We recruited 403 participants from the United Kingdom on Prolific (www.prolific.com), who received £1 in exchange for participation. We recruited a large sample to increase certainty in our posterior distributions. Based on preregistered criteria, we removed 17 participants who gave an irrelevant or nonsensical description of the emotion goal, three participants who did not remember their goal on the comprehension check, and 27 participants who recalled the incorrect goal condition on the comprehension check (these criteria are identical to those in Experiments 1a and 1b). We also preregistered that we would remove participants who made choices in under 1 s, but there were no participants meeting this criterion in the sample. Thus, our final sample was 356 participants (M_{age} = 37.50, SD_{age} = 19.44, 262 women, 91 men, three who preferred not to specify their gender).

Design

We employed a 2 (emotion expression goal: express vs. not express) × 2 (emotion experience goal: experience vs. not experience) between-subjects design.

Procedure and Materials

Emotion Goal Manipulation. The emotion goal manipulation was identical to Experiment 1a.

Emotion Regulation Strategy Choice. Participants were told that for the rest of the study, we would ask them to think about the situation they just described. We reminded participants of their goal and asked them to think about which strategies may be effective in achieving their goal. We focused on the same six emotion regulation strategies included in the earlier experiments.

We told participants that they would see 10 questions. For each question, they would be presented with three of the six strategies they could have used to achieve their goal. For each of these questions, they were asked to choose the strategy that would be *least effective* in meeting their goal, and *most effective* in meeting their goal from the three strategies provided.

Participants then completed a practice trial in which they were given dummy information so they could learn the question format. They could not progress beyond this practice trial until they answered the question as instructed. Then, they completed the 10 trials of the Discrete Choice Experiment. For each of the 10 trials, the prompt question was "Which of the following strategies do you think would be *least effective* and *most effective* in meeting your goal to [goal condition prompt]?" Figure 3 depicts the question format.

Comprehension Check. This was designed to identify participants who could not recall the goal to which they were assigned. The item was "During the study, what was the goal we asked you to select strategies for?" There were four response options corresponding to the four conditions, as well as a "do not remember/do not know" option. Participants were excluded from analysis if they could not correctly identify their condition on this item.

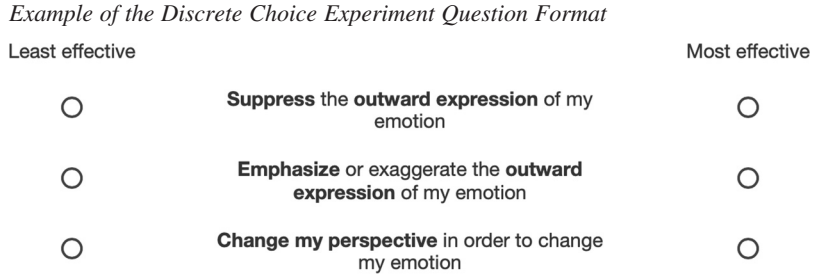
Analysis

The MNL analysis for the repeated-choice task in Experiment 3 was a modified version of the MNL analysis in Experiments 1 and 2. This modification enhances the power, although also the complexity, of the model. We restrict our focus in the main text to the utility parameters, which can be interpreted in the same manner as Experiments 1 and 2. Supplementary materials contains complete details of the MNL model for Experiment 3. The primary difference is that the repeated-choice task combined with the more complex MNL model substantially increases the precision of the utility estimates (see Figures 2 and 4), which translates to improved statistical power.

Results and Discussion

All analyses in Experiments 3a and 3b were conducted as specified in the preregistered analysis plan, except for one deviation and one extension. First, we deviated from our preregistered

Figure 3
Example of the Discrete Choice Experiment Question Format



Note. Bolding emphasizes the strategy for participants.

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analytic strategy, which originally stated that the participant-level utility estimates would be analyzed with a 2 (experience vs not experience) × 2 (express vs not express) between-subjects ANOVA. Conceptually, this analysis looks *within a given strategy*, and asks the question of which goal leads to this strategy being chosen most often. However, our analytic strategy in the previous studies looks *within a given goal*, and asks the question of which strategy is chosen most to meet that goal. Choosing this more theoretically relevant comparison, and mirroring the analyses in Experiments 1 and 2, we changed our preregistered within *strategy* comparison to a within *goal* comparison. Second, we extended our analytic strategy to include an analysis of participant-level utility estimates to assess the overall utility of each strategy independent of goal condition. While not mentioned in our original preregistration, we felt it was a theoretically relevant question to ask which strategy was chosen most overall.

Strategy Utility

Our analysis accounts for differences in the base rate use of each strategy: That is, it accounts for the fact that participants have a general preference to use some strategies and to avoid others. Differences in base rates mean that some strategies were chosen more often than others independent of the manipulation of the four goal conditions. A consequence is that the results of Experiments 1 and 2 potentially aggregate two effects: the goal manipulation effect, which is pertinent to the primary hypotheses, and the base rate preference for each strategy, which is interesting in itself but not pertinent to the primary hypotheses. The two effects can be separated in Experiment 3, as we have utility estimates for each participant, which were estimable because each participant contributed multiple data points (i.e., 10 best choices and 10 worst choices). To this end, we calculated the grand mean utility across *all* participants separately for each of the six strategies (i.e., we marginalized, or averaged, over goal condition). These are the values shown in Figure 4.

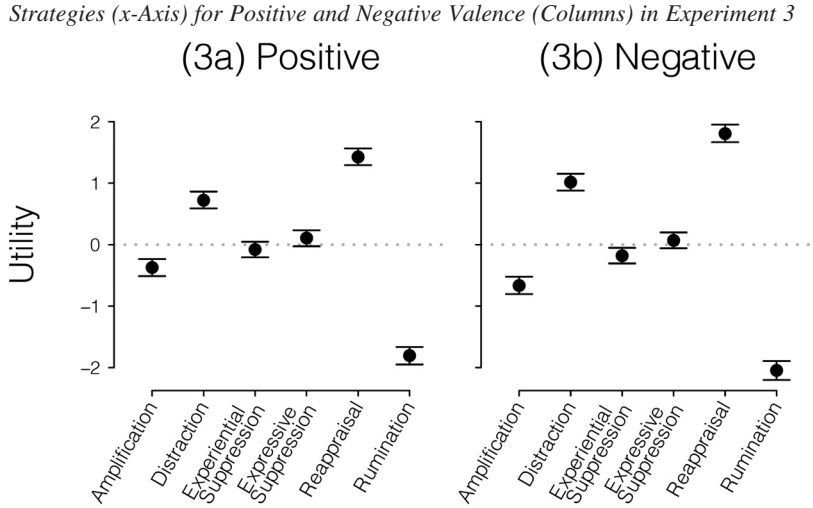
Inspecting the grand mean utility across all strategies reveals that reappraisal and distraction had *positive* utility estimates and rumination and amplification had *negative* utility estimates. This indicates that participants had a bias toward choosing reappraisal and distraction, and a bias away from choosing rumination and amplification across all four goal conditions. In contrast, experiential and expressive suppression were neutral, suggesting that participants neither actively preferred nor avoided these strategies.

Goal Utility

To control for base rate differences in the preference for different strategies (i.e., Figure 4), we subtracted the grand mean utility from the corresponding utility parameter for each participant, separately for each of the six strategies. This process is conceptually similar to ANOVA; the grand mean of the dependent variables is subtracted from all scores to obtain clearer estimates of condition effects. We then aggregated the individual-participant utility parameters to investigate between-condition differences in utility for the six strategies. That is, for every sample from the posterior distribution, the utility estimates for each strategy were averaged across the participants in each condition. This generates the posterior distribution of the mean participant effect, which is summarized by 95% credible intervals in Figure 2. This is conceptually the same procedure as averaging raw data across participants in a condition; the difference is that the procedure is applied recursively for every sample from the posterior distribution of the parameters.

For Goal 1 (experience and express emotion), rumination was chosen more often than other strategies but *less* often than amplification. For Goal 2 (experience but not express emotion), expressive suppression was chosen more often than other strategies. For Goal 3 (express but not experience emotion), amplification was

Figure 4
Posterior Distribution of the Utility Parameters for the Six Emotion Regulation Strategies (x-Axis) for Positive and Negative Valence (Columns) in Experiment 3



Note. Utility is marginalized over goal conditions. Dots and bars show the posterior median and 95% credible interval, respectively. The dotted gray line is a reference point showing zero utility (neither positive or negative utility). Positive utility indicates a strategy is more likely to be chosen and negative utility indicates a strategy is less likely to be chosen.

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chosen more often than other strategies except experiential suppression and distraction. For Goal 4 (neither experience nor express emotion), distraction was chosen more often than any other strategy, but reappraisal was chosen *less* often than rumination and experiential suppression.

As in most of the previous experiments, expressive suppression was chosen most often to meet Goal 2 to experience but not express emotion. A clearly preferred strategy did not emerge for the other asymmetric goal, Goal 3—to experience but not express emotion. As in past experiments, participants selected amplification and rumination to meet Goal 1—to experience and express emotion—although with added the discrimination possible with our new analytic approach we found amplification was chosen even more than rumination. In this experiment, distraction emerged as a preferred strategy selection to meet Goal 4—to neither experience nor express emotion—and unexpectedly reappraisal was *less* preferred than some other strategies to meet this goal, even though it is effective in downregulating emotion experience and expression (Webb et al., 2012).

Experiment 3b: Negative Emotion

Method

This study was preregistered: https://aspredicted.org/py2ua.pdf.

Participants

We recruited 401 participants from the United Kingdom using the online platform Prolific, who received £1 in exchange for their participation. Based on our preregistered criteria, we removed 21 participants who had an irrelevant or nonsensical description in the emotion goal manipulation, three participants who reported not remembering their goal on the comprehension check, and 40 participants who recalled the incorrect goal condition on the comprehension check. We also preregistered that we would remove participants who made choices in under 1 s, but there were no participants meeting this criterion in the sample. Thus, our final sample was 337 participants (M_{age} = 39.47, SD_{age} = 12.73, 245 women, 90 men, one nonbinary, one other).

Design, Procedure, Materials, and Analysis

These were identical to Experiment 3a, except all references to positive emotion were replaced with negative emotion.

Results and Discussion

Strategy Utility

Inspecting the grand mean utility across all strategies (see Figure 4) again revealed a bias toward choosing reappraisal and distraction, and a bias away from choosing rumination and amplification across all four goal conditions. In contrast, experiential and expressive suppression were relatively neutral.

Goal Utility

Results showed that for Goal 1 (experience and express emotion), rumination was chosen more often than any other strategy. For Goal 2 (experience but not express emotion), expressive suppression was chosen more often than any other strategy except distraction. For Goal 3 (express but not experience emotion), amplification was chosen more often than any other strategy. For Goal 4 (neither experience nor express emotion), distraction and reappraisal were chosen more often than any other strategy.

Results differed slightly from Experiment 3a, possibly due to the change in valence. This time, rumination was most preferred for Goal 1—to experience and express emotion, and amplification was most preferred for Goal 3—to experience but not express emotion. Participants were divided between expressive suppression and distraction to meet Goal 2—to experience but not express emotion. For the first time, reappraisal and distraction were the most preferred strategies to Goal 4—to neither experience nor express emotion.

Results Summary

Across experiments, certain goals prompted relatively regular choice of emotion regulation strategies, while other goals were not consistently associated with strategy choices. Generally speaking, where consistencies occurred these tended to line up with strategies theory would predict to be relatively effective in meeting a given emotion goal. Table 2 summarizes the most chosen strategy as a function of goal condition.

Goal 1: Experience and Express Emotion

As rumination increases felt and shown emotion, it could be used to meet a goal to experience and express emotion. Consistent with this logic, people chose rumination to meet this goal more than any other strategy, excepting amplification. Amplification was sometimes chosen at similar rates, and in two experiments (both positive emotion) was chosen *more* than rumination. As a strategy that targets expression but leaves experience unchanged, amplification would be effective in targeting the expressive dimension of Goal 1 but is not guaranteed to target felt emotion in desired ways.

Both rumination and amplification had positive goal utility estimates for the goal to experience and express emotion across all experiments (see Figure 2). This preference is even more notable in light of the evidence from the overall utility estimates (see Figure 4) that these strategies were less likely to be chosen in general when averaging across all goal conditions.

Goal 2: Experience but Not Express Emotion

We hypothesized that expressive suppression would be a useful strategy to meet the goal to experience but not express emotion, because it targets emotion expression and appears to leave emotion experience unchanged. Consistent with this expectation, expressive suppression had a positive utility estimate for Goal 2 in all experiments (see Figure 2). This strategy was also chosen more consistently than any other strategy *except* distraction, which was chosen equally often in four of the six experiments (three negative emotion and one positive emotion). Although expressive suppression was generally not preferred overall (neither was it particularly avoided; Figure 4), it was still mostly chosen to meet this goal across the experiments.

Table 2
Summary Across Experiments of Most Chosen Strategy as a Function of Emotion Goal

Experiment	Goal 1 Experience AND Express	Goal 2 Experience NOT Express	Goal 3 Express NOT Experience	Goal 4 Neither Experience NOR Express
Experiment 1				
(a) Positive	Amplification	**Expressive suppression**	**Amplification** Experiential suppression	**Distraction** **Reappraisal** Expressive suppression Experiential suppression
(b) Negative	**Rumination** Amplification	**Expressive suppression** Distraction	None	**Distraction**
Experiment 2				
(a) Positive	**Rumination** Amplification	**Expressive suppression** Distraction	**Amplification**	**Distraction** Expressive suppression
(b) Negative	**Rumination**	**Expressive suppression** Distraction Reappraisal Rumination	**Amplification** Experiential suppression Distraction Reappraisal	**Distraction**
Experiment 3				
(a) Positive	Amplification	**Expressive suppression**	**Amplification** Experiential suppression Distraction	**Distraction**
(b) Negative	**Rumination**	**Expressive suppression** Distraction	**Amplification**	**Distraction** **Reappraisal**

Note. Bold indicates strategies that were consistent with theoretical expectations.

Goal 3: Express but Not Experience Emotion

Given it targets emotion expression but appears to leave experience unchanged, we hypothesized that amplification would be able to meet the goal to express but not experience emotion. There was some evidence for this: Amplification was the most chosen strategy to meet this goal in three of the six experiments (two positive emotion and one negative emotion). The goal utility estimates for this strategy in Goal 3, although all positive (see Figure 2), were relatively small, even Experiments 3a and 3b—the studies that accounted for individual differences in choice pattern. As such, conclusions about choices regarding this strategy should be interpreted with caution.

Goal 4: Neither Experience Nor Express Emotion

For the goal to neither experience nor express emotion, we hypothesized that distraction and reappraisal would both be effective. Although reappraisal was the most consistently preferred strategy overall (see Figure 4), there was little evidence that it was differentially preferred in this particular goal context. Indeed, the goal utility estimate for reappraisal was not consistently positive in the Goal 4 condition across all experiments (see Figure 2). Distraction, on the other hand, did have a consistently positive goal utility estimate, and was preferentially chosen to meet this goal relative to other strategies in four of the six experiments (three negative emotion and one positive emotion). This pattern was particularly pronounced in the latter two experiments that accounted for individual differences in choice patterns.

General Discussion

This is the first empirical research to investigate emotion goals that bring emotion experience and expression into or out of alignment. It is also among the first experimental work to extend emotion regulation strategy choice paradigms beyond two strategies at a time, investigating choice between six common emotion regulation strategies. By offering a wider variety in choice, we better approximate the kind of strategy selection options that are available to people in daily life. Our methods allowed us to determine which strategies were selected to meet specific goals, and later experiments used a choice paradigm drawn from cognitive psychology to reveal and account for general preferences toward (and avoidance of) certain strategies in making these determinations. We found that certain emotion goals prompt selection of strategies that should be effective according to theory, notably goals that prioritize down-regulation of emotion expression (Goals 2 and 4), while others do not.

Our findings are somewhat consistent with past research that has investigated goals for emotion experience, while also further contextualizing these past findings. For example, Eldesouky and English (2019) found that prohedonic goals to decrease negative emotion were associated with greater self-reported use of reappraisal and distraction. Similarly, Millgram et al. (2019) found a preference for distraction over rumination when participants were experimentally induced to have a goal to reduce negative and positive emotion experience. We found participants consistently selected distraction, not reappraisal, when they had a goal to decrease experienced emotion, but critically only when participants also had a goal to decrease expressed emotion (i.e., Goal 4—to neither experience nor express emotion). In fact, when participants had a goal to *experience but not express* negative emotion (Goal 2 in Experiments 1b, 2b, and 3b), they sometimes chose

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distraction at rates similar to expressive suppression. This suggests that work investigating emotion experience goals may risk conflating goals for emotion experience and expression, which speaks to the value of assessing both goals when investigating their impact on strategy selection.

Millgram et al. (2019) also assessed strategy choice when participants had a goal to increase emotion experience, finding under these conditions rumination was a preferred strategy relative to distraction. We found support for the notion that people choose rumination when they want to increase experience, but only when they had a simultaneous goal to increase expressed emotion (i.e., Goal 1—to experience and express emotion). When participants had a goal to experience but not express emotion (Goal 2), expressive suppression was mostly preferred over rumination (with the exception of Experiment 2b). These findings further highlight the benefits of assessing emotion expression goals alongside emotion experience goals, as well the value of assessing choice between a wider range of emotion regulation strategies to model variation in choice between a wider array of options.

Of course, it is difficult to directly compare our findings against past work on emotion experience goals, because adding the expressive dimension may alter the nature of the emotion experience goal. In other words, a goal to decrease emotion experience may be qualitatively different and held for different reasons than a goal to decrease emotion experience *while also* decreasing emotion expression. Future work should investigate this empirically by assessing choice under conditions of no goal, "one-dimensional" emotion goals (i.e., focusing solely on either experience or expression) and "two-dimensional" emotion goals (i.e., focusing on both simultaneously; Greenaway & Kalokerinos, 2019), to investigate potential differences in their impact on strategy selection. In addition, future research could explore asymmetries in the relative intensity with which people hold goals for emotion expression versus emotion experience. In other words, research could explore the degree to which goals that prioritize expression over experience may yield different emotion regulation choices (e.g., suppression) compared with goals that prioritize emotion experience over expression (e.g., distraction).

Theoretical Implications

In addition to contextualizing past work on emotion experience goals, our research has implications for emotion regulation theory as construed at all stages of the extended process model (Gross, 2015). Given that much of the existing emotion regulation literature has focused on the consequences of emotion regulation strategies, we know comparatively less about the antecedents of these strategies. The emotion goals literature has been instrumental in understanding the identification stage of emotion regulation by revealing factors that prompt emotion regulation decisions. Our work contributes to this nascent area of research, revealing that it is not only how people want to *feel*, but also how they want to be *seen* to feel, that influences their decision to regulate.

Strategy Selection

More directly, our findings have implications for the selection stage of emotion regulation as we assessed emotion regulation strategy choice as a function of goal condition. Importantly, the range of strategies our participants were able to choose from outstripped the number in existing experimental paradigms, which have typically assessed choice between two strategies (e.g., Millgram et al., 2019; Sheppes et al., 2011; Sheppes et al., 2014). This approach allowed us to approximate more naturalistic forms of emotion regulation, where people choose from many strategies rather than making a dichotomous decision (Ford et al., 2019). Our findings revealed quite a bit of variation in strategy choice across goals and studies, hinting at the value of providing a broader choice set when asking people to select an emotion regulation strategy. This variation in strategy choice further indicates that people may be selecting strategies that match their context in ways predicted by regulatory flexibility theorizing (e.g., Bonanno & Burton, 2013).

Our use of the discrete choice paradigm in Experiments 3a and 3b enabled us to obtain an overall ranking of strategy preference, which yields new insight into how people compare strategies against one another (see Figure 4). Reappraisal was the most preferred strategy overall for regulating both positive and negative emotion, regardless of emotion goal. In some ways this is surprising, given that people typically use reappraisal little in everyday life (e.g., Brans et al., 2013) and find it difficult to use when instructed to experimentally (e.g., Kalokerinos et al., 2015). This may suggest people have difficulty using reappraisal, but believe the strategy will be useful and so are more likely to select it to meet a variety of emotion goals. Indeed, research shows that reappraisal is a versatile strategy that can be implemented in a variety of different ways (Jamieson et al., 2013; McRae et al., 2012).

The discrete choice paradigm also allowed us to distinguish the strategies people *avoid* in addition to the strategies they select. Results revealed that rumination was by far the most avoided strategy overall for regulating negative emotion (Experiment 3b; Figure 4). This makes sense, given ruminating on negative emotion is associated with a host of detrimental outcomes, such as depression and anxiety (e.g., Nolen-Hoeksema, 2000). What is perhaps more surprising is that this strategy was also avoided in the context of regulating positive emotion (Experiment 3a; Figure 4). Rumination functions to up-regulate emotion experience and expression, and thus has the potential benefit of *increasing* positive emotion in service of a prohedonic motive (e.g., Bryant & Veroff, 2007; Tamir, 2016). Consistent with this notion, rumination had a positive goal utility estimate (i.e., was more likely to be selected relative to zero; Figure 2) for Goals 1 and 3 in which people wanted to increase emotion experience, and had a negative goal utility estimate (i.e., was less likely to be selected relative to zero; Figure 2) for Goals 2 and 4 in which people wanted to decrease emotion experience. Thus, much like expressive suppression in the context of down-regulating emotion expression, our findings regarding rumination provide insight into situations in which people may choose to use traditionally maladaptive emotion regulation strategies because they believe these strategies will be useful in meeting a particular goal (e.g., Greenaway & Kalokerinos, 2017; Kalokerinos et al., 2017).

The possibility that people's beliefs about the utility or appropriateness of a given emotion regulation strategy may influence its selection aligns with an emerging literature on emotion beliefs (e.g., Ford & Gross, 2019). This literature shows that people's beliefs about emotions—for instance, whether emotions are good

versus bad, or controllable versus uncontrollable—shapes how they experience those emotions and how they regulate those emotions. This opens the possibility that just as people hold beliefs about certain emotions, they may hold beliefs about certain emotion regulation strategies. Such beliefs may concern the effectiveness of a given strategy in changing emotion in desired ways (e.g., ruminating only makes me feel worse in the long run), and the acceptability of using a given strategy (e.g., it is more socially appropriate to express than suppress emotion). Given our experiments, with the exception of Experiment 2a and 2b, relied on retrospective indication of the strategies people chose to meet a particular goal, it is possible that emotion regulation beliefs of this sort influenced recalled strategy choice. A fruitful avenue for future research would be to assess emotion regulation beliefs directly, and investigate how these beliefs guide the relationship between emotion goals and strategy selection. More broadly, this direction would gel with work showing the value of taking a lay person perspective to inform theories of emotion regulation (e.g., Parkinson & Totterdell, 1999).

Strategy Implementation

We approached the aim of mapping "two-dimensional" emotion goals to strategy choice in two ways: first by predicting which would be the most effective strategy to choose per goal based on empirical evidence and theory, and second by revealing actual strategy choice by participants. Some goals, such as those that involved down-regulation of emotion expression (Goals 2 and 4) revealed a relatively close mapping, such that the strategies theory would predict are likely to be effective were also the strategies most likely to be chosen (i.e., expressive suppression and distraction). Other goals, notably Goal 3 to express but not experience emotion, showed less correspondence between the predicted "most useful" strategy (amplification) and participant choice. This may suggest that people have relatively strong understanding of how certain strategies affect emotion experience and expression (e.g., expressive suppression), but less understanding of others (e.g., amplification). Alternatively, it may be that a goal to express but not experience emotion is less common than other types of two-dimensional goals, and that people thus have less experience choosing and effectively implementing strategies to meet this goal (see Greenaway & Kalokerinos, 2019 for a description of the contexts in which participants report holding each two-dimensional goal).

Noting areas of difference between strategies that theory suggests people *should* choose and those they *do* choose could provide fodder for developing interventions that promote effective emotion regulation. For example, distraction had positive goal utility (i.e., was more likely to be selected) for Goal 2—to experience but not express emotion—across all studies. This suggests some people may believe that distraction targets emotion expression but not experience, while research shows it affects both emotion dimensions (Webb et al., 2012). Misconceptions of this sort matter because if individuals do or do not believe a given strategy will meet a particular emotion goal, this belief may interfere with how much effort they put into strategy implementation (e.g., Molden & Dweck, 2006). Potential interventions could target understanding of certain emotion regulation strategies and make participants aware that particular strategies are more or less effective for achieving certain emotion goals.

Limitations, Strengths, and Future Directions

As with all research, our studies suffered from some limitations that temper the conclusions we can draw. Notably, statistical power was compromised in the early experiments—a weakness that was offset by our Bayesian analysis strategy as well as the larger sample size and use of a within-subjects discrete choice paradigm in later experiments.

In addition, our experiments were conducted in contexts that involved no social interaction between participants. This could mean that our results downplay the importance of goals for emotion expression, as expression is particularly important in the presence of other people (e.g., Sels et al., 2021; Williams et al., 2021). This was addressed somewhat in Experiments 2a and 2b, which incorporated an implicit social element by recording participants' facial expressions, although this is by no means a prototypical form of social interaction. Accordingly, future research should manipulate two-dimensional emotion goals within social interactions to better understand their effects.

Further limiting the clarity of the results, our lab-based experimental goal manipulations in Experiments 2a and 2b did not always affect emotion experience in expected ways, suggesting that participants did not always reach their desired emotional state. For example, assigning participants a goal to express (vs. not express) positive emotion in Experiment 2a also increased self-reported expressed positive emotion, although this effect was not present in Experiment 2b. Indeed, in Experiment 2b, even assigning participants a goal to experience (vs. not experience) negative emotion did not affect self-reported experienced negative emotion. This makes it difficult to disentangle what emotional state people *aimed* to achieve from the emotional state they *did* achieve. However, the fact that one of these experiments at least contained an emotion experience goal that was met by participants (Experiment 2a) increases our confidence when considering their findings together and in concert with those from Experiments 1 and 3.

Another potential limitation is that we constrained participants to choose only one strategy, rather than allowing participants to select all strategies that they may find effective. This is a limitation shared by other experiments on emotion regulation strategy choice. This weakness was addressed, although not fully resolved, in Experiments 3a and 3b that used a within-subjects design, which allowed participants to vary their choice of strategy across trials. An emerging research agenda acknowledges that people may select and implement multiple emotion regulation strategies at once; a concept referred to as *polyregulation* (Ford et al., 2019). Investigating coactivation of multiple strategies in response to specific emotion goals is an exciting direction for future research.

These limitations are balanced by the strength of our interdisciplinary methods and the fact that we used a variety of experimental designs ranging from retrospective recall of real emotion goals to lab-based manipulations. Augmenting these approaches, future work would benefit from investigating "two-dimensional" emotion goals using experience sampling methods that capture the richness of and variation in emotion goals in the context of

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daily life. A theoretical advancement would see future research investigate mechanisms through which people choose specific strategies to meet particular emotion goals. Moreover, research should begin to investigate instances of slippage between strategy selection and strategy implementation, testing whether strategies that people select to meet particular goals are in fact more effective in meeting that goal than strategies they do not select, or even avoid.

Conclusion

It is hard to reach a specific destination without a map as a guide. So too, it is hard to study effective emotion regulation without first understanding how people *want* to feel. This is the message of the emerging literature on emotion goals, and we have advanced this literature by studying emotion experience goals and emotion expression goals in concert for the first time. Our findings indicate that goals to experience and express emotion affect the strategies people choose to influence their emotional state. In particular, they highlight potential areas of difference between "one-dimensional" goals for emotion experience explored in past research and goals that add an expressive dimension. With this two-dimensional map, researchers can begin to explore new ways of understanding emotion regulation antecedents, processes, and consequences.

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If you are interested in reviewing manuscripts, please write APA Journals at Reviewers@apa.org. Please note the following important points:

- To be selected as a reviewer, you must have published articles in peer-reviewed journals. The experience of publishing provides a reviewer with the basis for preparing a thorough, objective review.
- To be selected, it is critical to be a regular reader of the five to six empirical journals that are most central to the area or journal for which you would like to review. Current knowledge of recently published research provides a reviewer with the knowledge base to evaluate a new submission within the context of existing research.
- To select the appropriate reviewers for each manuscript, the editor needs detailed information. Please include with your letter your vita. In the letter, please identify which APA journal(s) you are interested in, and describe your area of expertise. Be as specific as possible. For example, "social psychology" is not sufficient—you would need to specify "social cognition" or "attitude change" as well.
- Reviewing a manuscript takes time (1–4 hours per manuscript reviewed). If you are selected to review a manuscript, be prepared to invest the necessary time to evaluate the manuscript thoroughly.

APA now has an online video course that provides guidance in reviewing manuscripts. To learn more about the course and to access the video, visit http://www.apa.org/pubs/journals/resources/review-manuscript-ce-video.aspx.